

Unit 1 Introduction to electrostatics

1.1 Introduction to Electricity

The area of study that examines the phenomenon of electricity at rest is called **electro statics**. We know well that matters are made up of tiny particles such as electron, proton, etc. and particles are crucial for electricity.

The force that causes electro to revolve around the nucleus on the orbit or shell is called **Electromagnetic force**.

Electricity is created by the flow of electrons. The movement of electrons, especially free electron is what refers to Electricity.

1.2 History of Electricity

Electric cells were developed by **Alessandro Volta** in 1800. He used copper and zinc in opposite directions in salt water and observed chemical reactions created current.

Micheal Faraday found out that moving magnet through copper wire created a current; which later known as Micheal Faraday law of induction.

Thomas Alba Edison made bulb which age light during night using DC current. Later **Nicola Tesla** invented AC current which became the foundation for modern electrical technology.

Turbine driven generators are created in 1890 to produce electricity. The Niagara Falls hydroelectric power plants built by Tesla start producing electricity. Now today, we see more advancement of Electricity generation.

1.3 types of Electricity

There are two types of electricity.

- a. Static Electricity
- b. Dynamic Electricity

a. Static Electricity

Static electricity results from electrons separating from their atoms. The accumulation of an electric charge in one place is known as **static electricity**. The charge remains stationary and does not move. For an example, When we take comb and rub on hairs, The electron of hair gets transferred into comb And now if we take comb near paper pieces then it attracts paper pieces as it has excess number of electron than paper pieces (i.e. positively charged).

The Charge Represented by “**Q**” and its unit is **Coulombs (C)**.

b. Dynamic Electricity

In dynamic electricity, there is a constant flow of electrons from one point to another producing electric current. It travels over wires from a distance. The electricity that is produced by any electrical sources in electrical circuits is said to be as current electricity. Hence the electricity that we use in our home is perfect example of current electricity. All of the electrical devices we use, including phones, electric bulbs, Washing machines, etc. are powered by electric current.

Difference between Static and Dynamic electricity by yourself

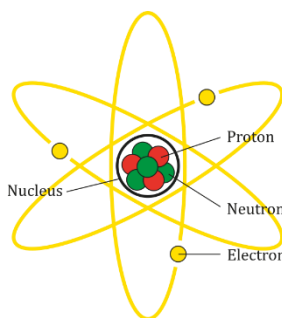
1.4 Application and uses of electricity

- a. Electricity in House
- b. Electricity in Entertainment
- c. Healthcare
- d. Transportation
- e. Industries and factories

1.5 Atom and its component

The element that is found around is made up atoms. **Atoms** are the smallest unit of an matter or element that inherits the characteristics of matter. **Atoms** are made of three components. They are **proton, neutron, and electron**.

Proton is positively charged. **Electrons** are negatively charged, and **Neutrons** are neutral in charge.



1.6 Atomic number, Atomic Weight, Free Electrons, and Electric charge

Atomic Number:

Atomic number can be defined as the total number electrons or protons contained in an atom of element. It is represented by “Z”. Chemical elements can be uniquely identified by its atomic number.

Atomic number = No. of **protons** or No. of **electrons** contained in atom

For example, atomic number of oxygen is 8 as it consists of 8 protons **or** 8 electrons.

Atomic Weight

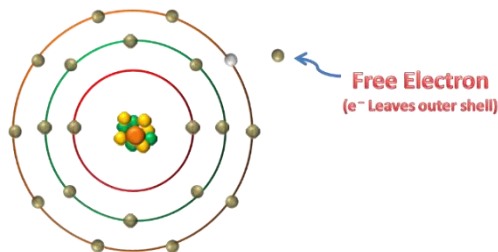
The total number of protons and electrons contained in an atom can be defined as atomic weight. The unit of atomic mass is **a.m.u** (atomic mass unit).

At.Weight = total number of neutrons + total number of proton

For example, the atomic weight of oxygen is 16 because it consist of 8 neutrons and 8 protons.

Free electrons

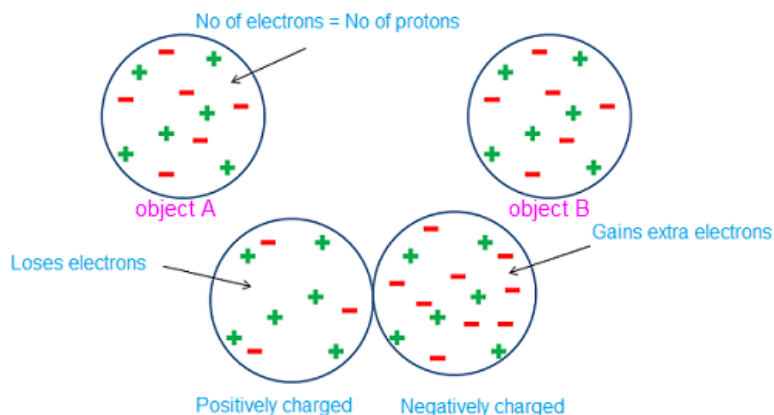
The electrons which are not bound to the nucleus, and which can freely move in response to external energy sources are known as **free electrons**. The term “electric current” means movement of electron within substance. The materials that **have free electrons** can conduct electric current and the materials which **do not have** free electrons cannot conduct electric current.



Electrical Charge

Electric charge is a fundamental property of matter that causes it to experience a force when placed in an electric field. An **electric field** is an invisible area around a charged object where it can **push or pull** other charges. positive and negative charge is generated by protons and electrons respectively. Electric charge is produced on atom as electron goes from one atom to another.

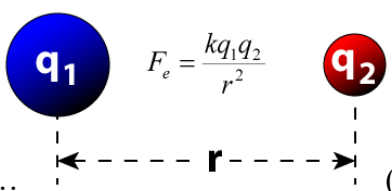
Atoms which lose electrons are **positively charged** and atoms which gain electrons are **negatively charged**.



1.7 Coulomb's law

Coulomb's law states that “the electrostatic force between any two charged body is directly proportional to the product of the magnitude of these charges and inversely proportional to the square of the distance between them”.

Let us consider two charges “ q_1 ” and “ q_2 ” is separated by distance “ r ” and “ F ” is the electrostatic force that creates attraction and repulsion between the charges.



Force(F) $\propto q_1 \cdot q_2$ (i)

Force(F) $\propto 1/r^2$ (ii)

Combining (i) and (ii)

$$F \propto q_1 \cdot q_2 / r^2$$

$$\therefore F = k q_1 \cdot q_2 / r^2$$

Where, k = proportionality constant (Columb's constant) = $9 \times 10^9 \text{ Nm}^2/\text{C}^2$

1.8 Introduction to Electric Field, Potential and Potential Difference

Electric Field

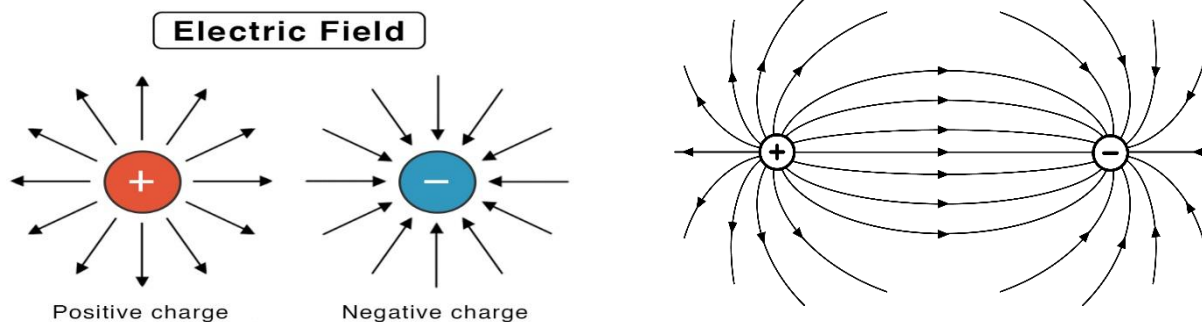


Fig: Electric field

An electric field is a physical field that surrounds electrically charged particles and acts as an attractor or repellent to all other charged particles in the vicinity.

Mathematically,

$$E = F/q$$

Hence, the strength of electric field “E” at any point may be defined as the electric force “F” exerted per unit positive charge “q” at that point.

Electric field are typically shown as lines of force or field lines, which begin on positive charges and end on negative charges.

Electric potential

The amount of energy that requires to move a unit of charge from a reference point to a specific point against an electric field is called electric potential. It is also known as electric potential, potential drop. The potential at reference point is zero unit.

Electric potential difference

Potential difference is **the difference in** the amount of energy that charge carriers have between two points in a **circuit**. The charge goes from higher potential to lower potential because there exists a potential difference between two points.

Mathematically,

$$V_A - V_B = \text{work to move charge } q \text{ from B To A} / \text{Charge } q$$

If **1 joule** of work is done **to move 1 coulomb charge** from point B to another point A with a circuit, then the potential difference between the points is said to be **1 volt**.

1.9 Electrical Energy, Voltage, and its Unit

Electrical energy

Electrical energy is the force that is produced from the transfer of electron from one location to another. The basic unit of electrical energy is the **joule** or **watt-second**.

Electrical energy is said to be 1 joule or **watt-second** when 1 ampere (1A) current flows through the circuit for 1 second and is 1 volt potential difference is applied across the circuit.

$$1\text{kWh} = 1000 \times 60 \times 60 \text{ watt-second}$$

Voltage

The force which moves free electron between the atoms is known as **voltage**. Battery or a generator is used to supply voltage. It is represented by symbol “V”. For example, 12V is smaller than 230V.